

engineers design systems, write code, create workflows, build models, and solve difficult technical problems. But when it comes to protecting these creations, especially in software- and algorithm-based innovation, the path is often unclear. Valuable work can slip into a grey zone where it is used, copied, relocated, or commercialised elsewhere, with little or no ownership remaining in India.

This is not a narrow legal detail. It changes behaviour. When ownership is uncertain, ambition shrinks. Investors hesitate. Founders look outward. Companies structure themselves abroad. Over time, the place where work is done and the

orchestration, fraud detection, recommendation engines, process automation, AI training pipelines, and platform intelligence are core assets that generate significant economic value. The lesson is clear: in today's world, economic power increasingly comes from owning underlying logic rather than supplying technical labour.

What this means for India

This is where India's challenge becomes visible. Under the current interpretation of Section 3(k), software and algorithms often lack a clear and enforceable path to protection. The result is predictable. Work may be done here, but ownership

how people think. It encourages builders to create for the long term. If India wants to move from implementation to creation, software- and algorithm-based innovation must have a clear and enforceable path to protection.

But ownership alone is not enough. Innovation creates economic power only when ideas move beyond theory into products that are tested, adopted, manufactured, or deployed, and scaled. That is the real bridge from classroom learning to industry, exports, and national prosperity.

The foundation of innovation

Every major breakthrough is built on repeated attempts. Innovation is not a one-time event. It is a process of trying, failing, learning, improving, and trying again with better judgment.

This is where India faces a deeper challenge. We have talent. We have ideas. But we do not yet have a broad culture that treats failure as part of progress rather than stigma. Too many students are trained to avoid mistakes rather than explore possibilities. Too many professionals choose safe paths because the social and economic cost of failure remains high. That shapes behaviour long before innovation has a fair chance to emerge.

Yet, every major innovation ecosystem works in the same way. More attempts create more learning. More learning improves the odds of success. Out of a large number of experiments, a few succeed at scale. Those successes are visible. The many failed or partial attempts behind them are usually invisible, but they are the true foundation.

India must become more comfortable with this reality. Students should be encouraged to build early, test ideas, create prototypes, and solve practical problems. A working model, even if imperfect, often teaches more than flawless

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place where value is owned begin to diverge. That is a costly gap for any country that wants to become a leading technology economy.

Code is capital

In the modern technology economy, value is increasingly created by logic. Search engines, recommendation systems, cloud platforms, payment networks, logistics engines, security systems, AI models, and enterprise software all depend on algorithms. These are not secondary elements supporting a product. In many cases, they are the product.

In simple terms, code is capital.

The world's most valuable technology companies are not built merely on the number of engineers they employ, they are built on the logic they own, protect, refine, and scale. Search-ranking logic, cloud

and long-term value often migrate elsewhere. India builds; others own.

That pattern has consequences far beyond patents. When innovators believe their most important creations cannot be securely protected, they begin to think smaller. They may limit ambition, avoid deep product building, or move legal and commercial structures outside the country. Over time, India risks becoming strong in execution but weak in value capture. That is not enough for a nation that wants to move from service strength to technology leadership.

Countries that dominate modern technology do not merely educate talent, they create systems in which talent can invent, own, commercialise, and scale. They protect hardware, software, process, and platform innovation. That shapes